

EDUCATION

PhD in Computer Science University of Massachusetts Amherst (UMass Amherst), MA, USA	Sep 2025
Master of Science in Computer Science University of Massachusetts Amherst (UMass Amherst), MA, USA	Feb 2024
Master of Science in Electrical and Electronic Engineering Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh	Jan 2021
Bachelor of Science in Electrical and Electronic Engineering Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh	Feb 2017

WORK EXPERIENCE

Applied Scientist II , Machine Learning Accelerate, Amazon.com Inc., Seattle, USA	Sep 2025 - Present
- Developing scalable pipelines for training and serving seller embeddings to support downstream ML applications - Investigating multimodal embedding fusion strategies to improve representation quality across heterogeneous seller signals	
Graduate Assistant , BioNLP Lab, UMass Amherst, MA, USA	Sep 2019 - Aug 2025
- Advisor: Professor <i>Hong Yu</i> - Areas of Study: Clinical Natural Language Processing, Epidemiology	
Applied Scientist Intern , Machine Learning Accelerate, Amazon.com Inc., CA, USA	Jun 2024 - Aug 2024
Project: Hallucination localization in LLM responses.	
Applied Scientist Intern , Alexa AI, Amazon.com Inc., WA, USA	May 2023 - Sep 2023
Project: Multimodal content generation from long-form text.	
Applied Scientist Intern , Alexa AI, Amazon.com Inc., WA, USA	May 2022 - Sep 2022
Project: Building a synthetic dataset for utterance similarity detection and assessing its feasibility.	
Applied Scientist Intern , Alexa AI, Amazon.com Inc., WA, USA	Jun 2021 - Aug 2021
Project: Hallucination mitigation for constrained contextual query rewriting (CQR).	
Machine Learning Researcher , Semion Ltd., Dhaka, Bangladesh	Mar 2017 - Feb 2019
- Delivered machine learning solutions for healthcare clients, translating domain requirements into deployable models - Re-implemented state-of-the-art methods and adapted them to proprietary datasets for in-house product development - Designed and developed supporting software infrastructure for experimentation, model development, and deployment	
Intern , Semion Ltd., Dhaka, Bangladesh	Aug 2016 - Dec 2016

SELECTED PUBLICATIONS

NLP / ML

- X. Yin, **A. Mitra**, Y. Zhou, K. Chen, H. Yu. “*Scalable Counterfactual Risk Estimation for Rare Events in Longitudinal Data*”, KDD, 2026
- Z. Yao, Y. Zhao, **A. Mitra**, D. A. Levy, E. Druhl, J. Tsai, H. Yu. “*SynthEHR-Eviction: Enhancing Eviction SDoH Detection with LLM-Augmented Synthetic EHR Data*”, npj Digital Medicine, 2026 
- Z. Yang, **A. Mitra**, W. Hu, D. Berlowitz, H. Yu. “*Predicting suicide death among veterans after psychiatric hospitalization using transformer based models with social determinants and NLP*”, Scientific Reports, 2025 
- A. Mitra**, Z. Yang, E. Druhl, R. Goodwin, H. Yu. “*Synth-SBDH: A Synthetic Dataset of Social and Behavioral Determinants of Health for Clinical Text*”, EMNLP, 2025 
- A. Mitra**, K. Chen, W. Liu, R.C. Kessler, H. Yu. “*Post-discharge Suicide Prediction Among US Veterans Using Natural Language Processing-enriched Social and Behavioral Determinants of Health*”, npj Mental Health Research, 2025 
- A. Mitra**, N. Gupta, C. Naik, A. Sethy, K. Bice, Z. Raeesy. “*Generating contextual images for long-form text*”, LREC-COLING, 2024 

7. P. Vashisht, A. Lodha, M. Maddipatla, Z. Yao, **A. Mitra**, Z. Yang, J. Wang, S. Kwon, H. Yu. “UMass-BioNLP at MEDIQA-M3G 2024: DermPrompt-A Systematic Exploration of Prompt Engineering with GPT-4V for Dermatological Diagnosis”, 6th Clinical Natural Language Processing Workshop, ACL, 2024
8. Z. Yang, **A. Mitra**, S. Kwon, H. Yu. “ClinicalMamba: A Generative Clinical Language Model on Longitudinal Clinical Notes”, 6th Clinical Natural Language Processing Workshop, ACL, 2024
9. Z. Yang, **A. Mitra**, W. Liu, D. Berlowitz, H. Yu. “TransformEHR: Transformer-based encoder-decoder generative model to enhance prediction of disease outcomes using electronic health records”, Nature Communications, 2023
10. J. Wang, Z. Yao, **A. Mitra**, S. Osebe, Z. Yang, H. Yu. “UMASS-BioNLP at MEDIQA-Chat 2023: Can LLMs generate high-quality synthetic note-oriented doctor-patient conversations?”, 5th Clinical Natural Language Processing Workshop, ACL, 2023
11. Z. Yang, S. Wang, B. P. S. Rawat, **A. Mitra**, H. Yu. “Knowledge Injected Prompt Based Fine-tuning for Multi-label Few-shot ICD Coding”, EMNLP Findings, 2022
12. H. Ahsan, E. Ohnuki, **A. Mitra**, H. Yu., “MIMIC-SBDH: A Dataset for Social and Behavioral Determinants of Health”, Machine Learning for Healthcare, 2021
13. **A. Mitra**, B. P. S. Rawat, D. McManus, H. Yu. “Relation Classification for Bleeding Events from Electronic Health Records: Exploration of Deep Learning Systems”, JMIR Medical Informatics, 2021
14. **A. Mitra**, B. P. S. Rawat, D. McManus, A. Kapoor, H. Yu. “Bleeding Entity Recognition in Electronic Health Records: A Comprehensive Analysis of End-to-End Systems”, AMIA Annual Symposium, 2020
15. **A. Mitra**, K. Ashraf. “Sepsis Prediction and Vital Signs Ranking in Intensive Care Unit Patients”, arXiv preprint arXiv:1812.06686, 2018

NLP in Epidemiology

1. C. Leon, M. Sung, J. Reisman, W. Liu, R. D. Kerns, K. Gordon, **A. Mitra**, S. Kwon, H. Yu, W. Becker, W. Li. “The association of neurocognitive symptoms with opioid overdose events among patients on long-term opioid therapy”, Substance Use & Addiction Journal, 2026
2. S. Sultana, F. Ouyang, J. Qian, Y. Zhang, **A. Mitra**, R. Pradhan, Y. Moradi, T. Pogoda, J. Reisman, H. Yu. “One-Year Buprenorphine Initiation and Its Association with SDOH Among US Justice-Involved Veterans”, JAMA Network Open, 2026
3. F. Ouyang, W. Hu, J. Reisman, T. K. Pogoda, K. F. Carlson, W. Liu, Y. Moradi, Y. Zhang, S. Han, S. Sultana, W. S. Jang, Z. Yao, **A. Mitra**, H. Yu. “Association between PTSD and health-related social needs in US Veterans: an NLP analysis using Veterans Health Administration Data”, Journal of Affective Disorders, 2025
4. C. Leon, M.L. Sung, J. I. Reisman, W. Liu, R.D. Kerns, K.S. Gordon, **A. Mitra**, S. Kwon, H. Yu, W.C. Becker, W. Li. “Occurrence of opioid related neurocognitive symptoms associated with long-term opioid therapy”, The Clinical Journal of Pain, 2025
5. M.L. Sung, C. Leon, J.I. Reisman, K.S. Gordon, R.D. Kerns, W. Li, W. Liu, **A. Mitra**, H. Yu, W.C. Becker. “Disparities in receipt of medications for opioid use disorder before and during the COVID-19 pandemic in the U.S. Veterans Health Administration”, Substance Use & Addiction Journal, 2024
6. **A. Mitra**, R. Pradhan, R. D. Melamed, K. Chen, D. C. Hoaglin, K. L. Tucker, J. I. Reisman, Z. Yang, W. Liu, J. Tsai, H. Yu. “Associations Between Natural Language Processing-Enriched Social Determinants of Health and Suicide Death Among US Veterans”, JAMA Network Open, 2023 (nominated for VA Health Services Research & Development Best Paper Award 2023)
7. **A. Mitra**, H. Ahsan, W. Li, W. Liu, R. D. Kerns, J. Tsai, W. Becker, D. A. Smelson, H. Yu. “Risk Factors Associated With Nonfatal Opioid Overdose Leading to Intensive Care Unit Admission: A Cross-sectional Study”, JMIR Medical Informatics, 2021
8. A. J. Rose, J. S. Lee, D. R. Berlowitz, W. Liu, **A. Mitra**, H. Yu. “Guideline-discordant dosing of direct-acting oral anticoagulants in the veterans health administration”, BMC Health Services Research, 2021

In Review

1. F. Ouyang, T. K. Pogoda, J. Reisman, R. Li, R. Pradhan,, Y. Moradi, J. Mez, W. Liu, Y. Zhang, S. Sultana, J. Qian, Z. Yao, **A. Mitra**, H. Yu. “The Joint Effect of Social Determinants of Health and Posttraumatic Stress Disorder on Mild Cognitive Impairment, Alzheimer Disease, and Alzheimer Disease-Related Dementias in US Veterans: An NLP Analysis Using Veterans Health Administration Data”, JAMA Neurology, 2026 (In review)
2. **A. Mitra**, R. Pradhan, K. S. Gordon, H. Yan, R. D. Kerns, W. C. Becker, H. Tu. “Associations Between Natural Language Processing Enriched Social Determinants of Health and Fatal Opioid Overdose among US Veterans”, American Journal of Public Health, 2026 (In review)
3. R. Pradhan, Y. Zhang, F. Ouyang, W. Hu, **A. Mitra**, W. Liu, J. Reisman, D. Berlowitz, J. Mez, J. Kaufman, H. Yu. “Associations Between Natural Language Processing-Enriched Social Determinants of Health and Alzheimers Disease and Related Dementias Among United States Veterans”, Nature Aging, 2026 (In review)
4. M. L. Sung*, A. Feibel*, J. I. Reisman, K. S. Gordon, C. Leon, R. D. Kerns, W. Li, W. Liu, **A. Mitra**, H. Yu, W. C. Becker. “Impact of COVID-19 pandemic on receipt of medications for opioid use disorder among older Veterans”. Addiction Science & Clinical Practice, 2026 (In review).

SKILLS

Programming Languages: Python, Java, R, MATLAB, C, C++, HTML
Machine Learning Frameworks: PyTorch, TensorFlow, Keras, scikit-learn
Other Expertise: L^AT_EX, Git, Microsoft Office, Weka, Arduino

ACHIEVEMENTS

Dean’s List Award, for academic excellence, BUET, Bangladesh	2013, 2017
2 nd Runner up, Inter-University Project Show, BUET, Bangladesh	2015
1 st Runner up, Inter School & College Science Festival, Rajuk College, Bangladesh	2010
National Level Scholarship, Bangladesh	2003,2006,2009,2011

TEACHING EXPERIENCE

Spring 2025	Guest Lecturer, COMP.5700: Methods in Data Science, UMass Lowell
Fall 2020, 2023, 2024	Guest Lecturer, COMP.5800: Topics in Computer Science, UMass Lowell
Fall 2019	Teaching Assistant, COMPSCI 121: Introduction to Computing

SERVICES

- **Reviewer**
 - Conferences: ACL, NAACL, AACL, EACL, EMNLP, ICLR, AAAI, IJCNLP-AAACL, AMIA
 - Medical Journals: Transactions on Information Systems, Nature Mental Health, JMIR, JMIR Medical Informatics, JMIR Medical Education, JMIR AI, JMIR Aging, JMIRx Med, JMIR Infodemiology, JMIR Formative Research, npj Mental Health Research, BMC Pulmonary Medicine, BMC Health Services Research, Frontiers in Bioengineering and Biotechnology, PLOS One, Substance Use and Addiction Journal, Journal of Affective Disorders Reports
- **MS Application Reviewer**, 2023, Manning College of Information and Computer Science, UMass Amherst
- **Mentor**
 - COMPSCI 696DS (Spring 24, Spring 25)
 - Independent Study Projects (Spring 23, Fall 23, Spring 24, Spring 25, Fall 25)